

RAKSHIT DESHPANDE

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(Open to Relocate)

PROFESSIONAL SUMMARY

Python Developer with **5+ years** of experience designing scalable **.NET systems** and quantum-classical computing solutions across healthcare and research domains. Skilled in **Python, C#, system architecture**, and **microservices** with deep expertise in **DICOM integrations**. Committed to building high-performance, secure, and interoperable healthcare software; **actively seeking full-time opportunities** in software engineering.

PROFESSIONAL EXPERIENCE

BOSTON SCIENTIFIC

Boston, US

Python Developer

Jan 2025 -Present

Applying expertise from medical imaging and interventional systems to develop interoperable, high-performance healthcare software at Boston Scientific

- Designed and developed **.NET-based** modules for clinical workflow and imaging data integration, bridging patient EMR records with diagnostic imaging systems (DICOM) to improve data flow and reduce manual reconciliation
- Learned to containerize applications using **Docker** and deploy them on **AWS EC2** instances, gaining hands-on experience with basic cloud deployment and storage using S3
- Improved **multithreading and memory management**, lowering peak memory usage by 30% and preventing OOM crashes
- Developed dynamic web applications using **JavaScript, HTML, CSS**, and modern frontend frameworks like **React.js**
- Designed and optimized **SQL-based storage** for imaging metadata and patient telemetry
- Implementing **secure API gateways and microservices** for exchanging radiology and monitoring data between hospital systems, following HIPAA and ISO 13485 security standards
- Contributing to the development of **real-time alerting and notification modules** for patient monitoring.
- Built and optimized **3D and volume data viewers** for radiology images using rendering pipelines similar to those developed for interventional CT, improving visualization responsiveness and user experience
- Performed detailed **memory and performance profiling** using Visual Studio / dotMemory; optimized object lifetimes, reduced memory fragmentation, and improved service
- Implemented a configurable **laser-control subsystem** allowing runtime switching between 4-laser and 2-laser modes; integrated with firmware APIs, updated calibration transforms

UNIVERSITY OF MASSACHUSETTS

North Dartmouth, US

Software Developer

Feb 2024 - Dec2024

Designed and implemented a quantum-classical computer proof-of-concept to accelerate analysis of medical images using D-Wave annealers for future enhancements in medical imaging workloads.

- Developed a **proof-of-concept (POC)** leveraging quantum annealing on the D-Wave Advantage QPU to accelerate complex optimization problems arising in CT image analysis pipelines
- Investigated **hybrid quantum-classical workflows** for CT data denoising, image segmentation, and reconstruction — demonstrating potential reductions in processing time for **iterative algorithms** such as conjugate gradient and compressed sensing
- Conducted experimental evaluations using the **D-Wave Ocean SDK**, integrating classical reconstruction algorithms (Python, NumPy, SciPy) with quantum solvers to benchmark hybrid quantum-classical performance
- Formulated **reconstruction sub-problems** (e.g., sparse matrix inversion, iterative optimization) into QUBO / Ising models, enabling partial offloading of computationally intensive tasks to the quantum annealer for faster

convergence

HARMAN CONNECTED SERVICES

Software Engineer

Client: Roche, US

Bengaluru, India

Aug 2022 - Aug 2023

Applied medical imaging and technical expertise to build low-latency .NET microservices and visualization workflows for real-time patient telemetry.

- **Developed scalable .NET microservices** for remote patient monitoring, enabling continuous ingestion and real-time analysis of patient vitals from IoT-connected medical devices; optimized serialization and data pipelines to achieve low-latency streaming and minimal data loss during live signal rendering workflows
- Implemented **secure device communication interfaces** (REST/gRPC, MQTT) and patient data ingestion flows compliant with healthcare data standards (HL7 / FHIR)
- Applied experience from **medical imaging (Siemens Healthineers)** to design robust, fail-safe data acquisition workflows — ensuring data integrity, redundancy, and recovery for patient telemetry
- Developed real-time visualization dashboards using **DICOM**-derived vitals data, highlighting critical information to help radiologists expedite interpretation

SIEMENS HEALTHINEERS

Software Developer

2022

Bengaluru, India

Aug 2019 - Jun

Enhanced end-to-end DICOM imaging workflows by optimizing .NET processing pipelines and improving reliability and code quality — resulting in faster clinical performance, reduced technical debt, and lower operational downtime.

- Improved **DICOM** image-processing pipeline implemented in **C# .NET**, increasing throughput by 25% and reducing end-to-end processing latency by **40%**, supporting faster clinical workflows across 200 hospitals
- Revamped automation test suite using testing frameworks such as **NUnit, Karma, Jasmine**; Enhanced test coverage by 20%, resulting in identifying issues in the product at the development stage
- Identified code smells, proactively played a vital role in improving the team's Technical Debt. Improved code quality with **SonarQube**, identifying key hotspots, reducing technical debt by 75% and lowering critical/major code smells by a large margin
- Noticed redundant manual follow-ups and drove initiative to automate Jenkins-based CI/CD pipeline workflows, containerized services with Docker, and introduced basic orchestration practices, cutting deployment times and manual release steps by 40%
- Reduced mean time to recovery (MTTR) by **20%** by adding structured logging, distributed traces, and memory tracing, enabling faster root-cause analysis
- Reworked legacy multithreaded image processing modules to improve memory management and concurrency, lowering peak memory usage by **30%** and eliminating intermittent **OOM crashes**
- Developed an internal effort-tracking and notification tool (WinForms + C#) that automated daily burn/save reminders and sprint logging; saved ~1 hour/sprint per engineer by automating routine reporting

TECHNICAL SKILLS

- **Languages:** Python, Java, C, C++, SQL, C# (.Net)
- **Web & APIs:** Angular, Flask, HTML, CSS, JavaScript/React, Rest APIs, Angular
- **Testing:** N Unit, Karma, Jasmine
- **Data Science/ML:** Scikit-learn, Pandas, NumPy, Natural Language Processing, Image Processing
- **Distributed & Big Data:** PySpark, Hadoop, MapReduce, Scala
- **Cloud & DevOps:** AWS, Docker, Kubernetes, Azure
- **Databases:** SQL, MongoDB

EDUCATION

UNIVERSITY OF MASSACHUSETTS
MS in Computer Science (GPA: 3.64/4)

North Dartmouth, US
May 2025

NATIONAL INSTITUTE OF ENGINEERING, MYSURU
B.E (Hons.) Information Science & Engineering

Mysuru, India
Sept 2020